

Require: SET reviews $\mathcal{R} = \{r_1, r_2, \dots, r_N\}$, each with raw rating y^{raw}

Ensure: Adjusted ratings $\hat{y}^{\text{adjusted}}$

```

1: // Stage 1: Data Preprocessing
2: for each review  $r \in \mathcal{R}$  do
3:   Clean text (encoding errors, abbreviations, profanity tags)
4: end for
5: // Stage 2: Clause Extraction
6: for each review  $r$  do
7:    $\mathcal{C}_r \leftarrow$  segment  $r$  into clauses using spaCy
8: end for
9: // Stage 3: Aspect-Term Categorization
10: for each clause  $c \in \mathcal{C}_r$  do
11:   Embed  $c$  using all-mpnet-base-v2
12:   Compute cosine similarity  $\mathbf{s}_c$  against topic embeddings
13:   if  $\max(\mathbf{s}_c) > \tau$  then
14:     Assign  $c$  to topic  $\hat{k}_c = \arg \max_k s_{c,k}$ 
15:   else
16:     Assign  $c$  to Miscellaneous
17:   end if
18: end for
19: // Stage 4: Pedagogical Density Calculation
20: for each review  $r$  do
21:   Compute  $D_{\text{misc}}, D_{\text{eff}}, D_{\text{fair}}, D_{\text{work}}$ 
22: end for
23: // Stage 5: Emotion Extraction and Aggregation
24: for each clause  $c$  do
25:   Extract emotion vector  $\mathbf{e}_c$  using RoBERTa-GoEmotions
26:   Remove Neutral component to obtain reduced vector  $\mathbf{e}'_c$ 
27: end for
28: for each topic  $k$  do
29:    $E_k \leftarrow \frac{1}{|\mathcal{C}_k|} \sum_{c \in \mathcal{C}_k} \mathbf{e}'_c$ 
30: end for
31: // Stage 6: Feature Construction
32: for each review  $r$  do
33:    $\mathbf{x}_r \leftarrow [E_{\text{eff}}, E_{\text{fair}}, E_{\text{work}}, E_{\text{misc}}, D_{\text{eff}}, D_{\text{fair}}, D_{\text{work}}, D_{\text{misc}}]$ 
34: end for
35: // Stage 7: CatBoost Regression
36: Train  $f_{\text{CB}}$  on  $X$  to predict  $y^{\text{raw}}$ 
37: // Stage 8: Validity-Weighted Attenuation
38: for each review  $r$  do
39:    $\zeta \leftarrow 1 - (D_{\text{misc}})^s$ 
40:    $\hat{E}_{\text{misc}}^{\text{weighted}} \leftarrow E_{\text{misc}} \cdot \zeta$ 
41:   Construct attenuated feature vector  $\hat{\mathbf{x}}_r$  replacing  $E_{\text{misc}}$  with  $\hat{E}_{\text{misc}}^{\text{weighted}}$ 
42:    $\hat{y}^{\text{attenuated}} \leftarrow f_{\text{CB}}(\hat{\mathbf{x}}_r)$ 
43:    $\hat{y}^{\text{baseline}} \leftarrow f_{\text{CB}}(\mathbf{x}_r)$ 
44:    $\hat{y}^{\text{adjusted}} \leftarrow y^{\text{raw}} + (\hat{y}^{\text{attenuated}} - \hat{y}^{\text{baseline}})$ 
45: end for
46: return  $\hat{y}^{\text{adjusted}}$ 

```